



Fully Printable Gravity Knife Fidget Toy



Ordinary Contraptions

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Summary

A fully printable 5-piece gravity knife that can be printed without supports



5.75 hrs



1 pcs



0.20 mm



0.40 mm



PLA



36 g



Creality
Ender 3 Pro

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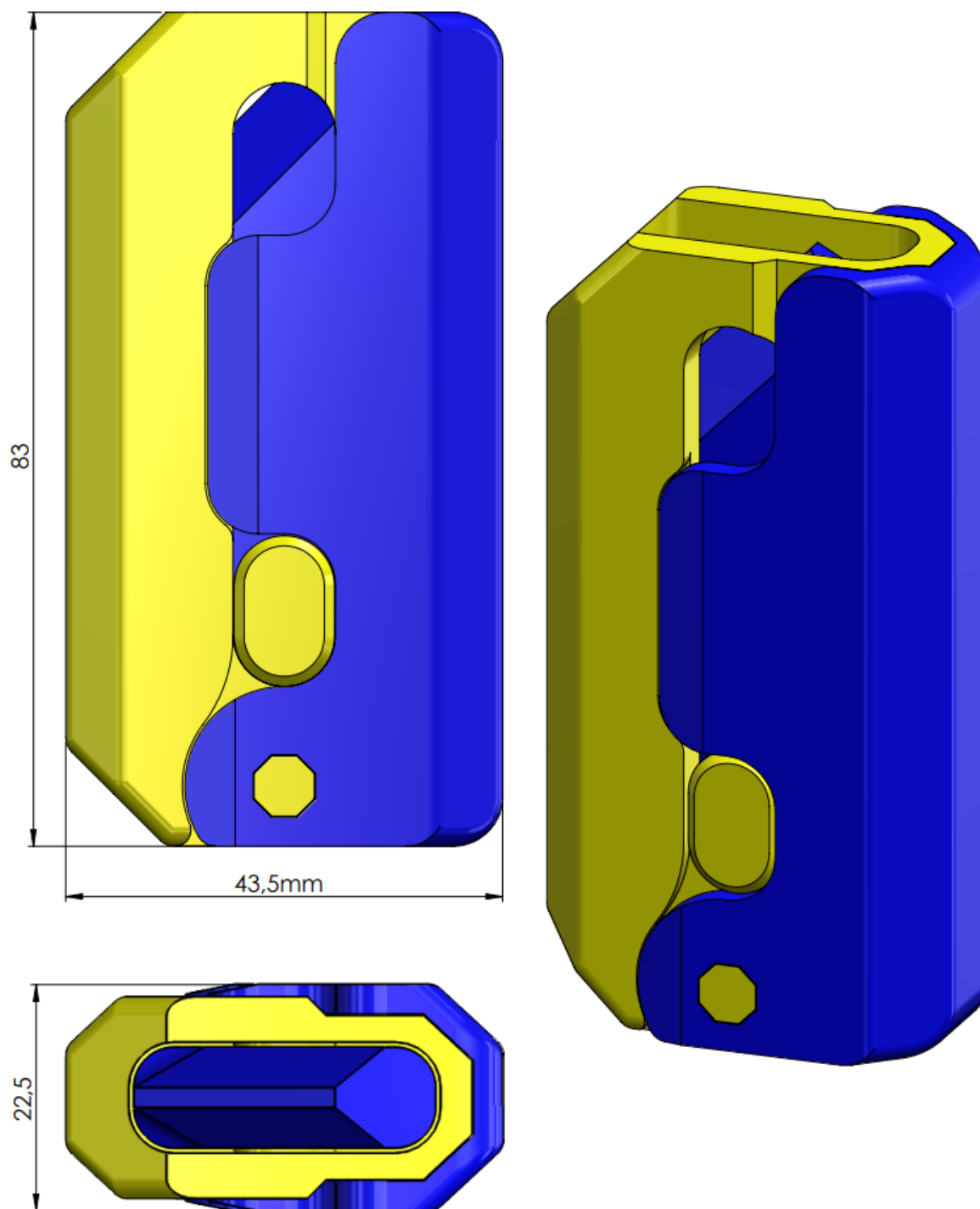
After taking heavy inspiration from other gravity knives desings, I decided to make my own fully printable version.

With this design you don't need any extra hadware, you can just print all the parts, even all at the same time if you want to.

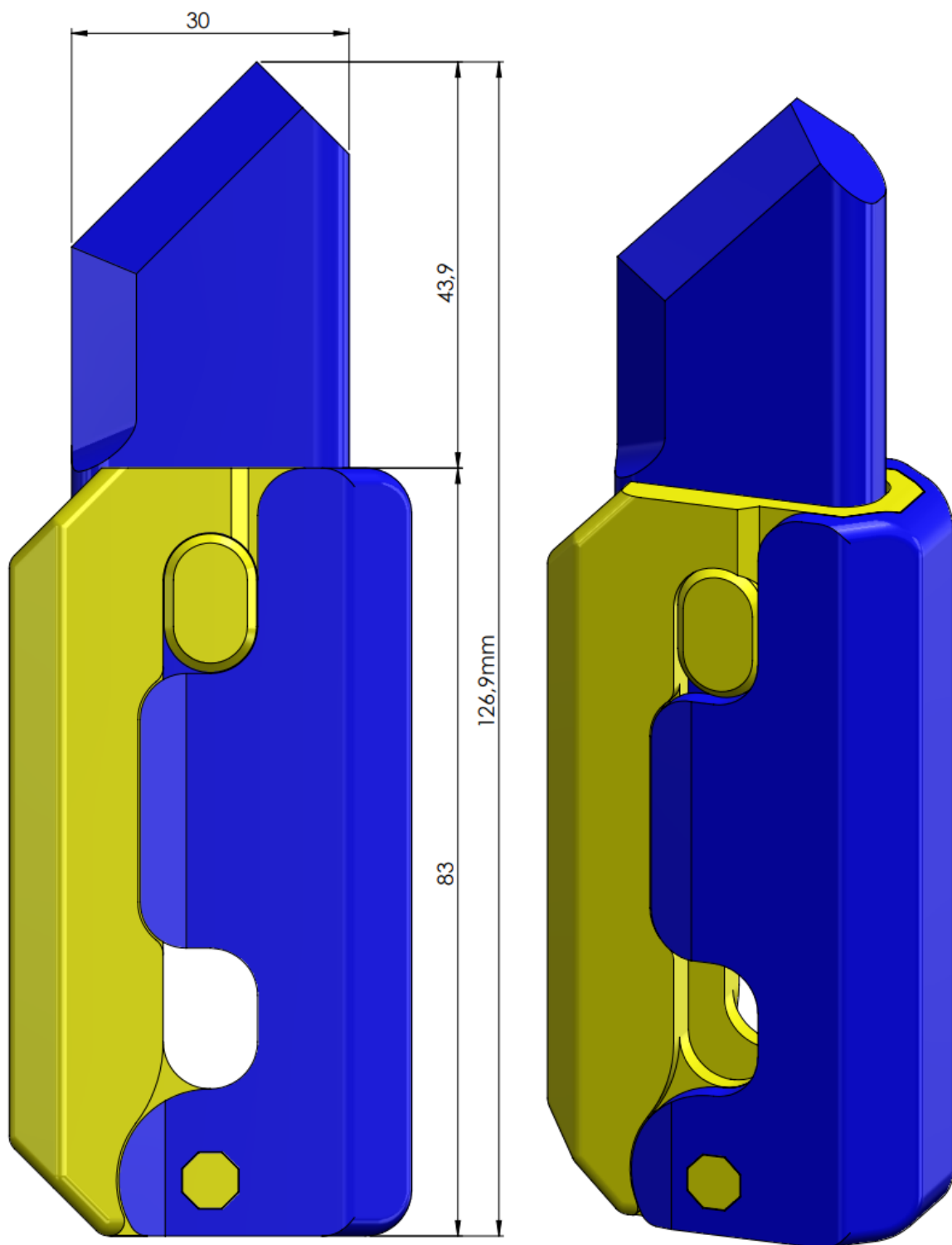
For a complete set you will need the following parts and quantities:

- x1 core
- x1 shaft
- x1 blade
- x1 guard
- x2 stops

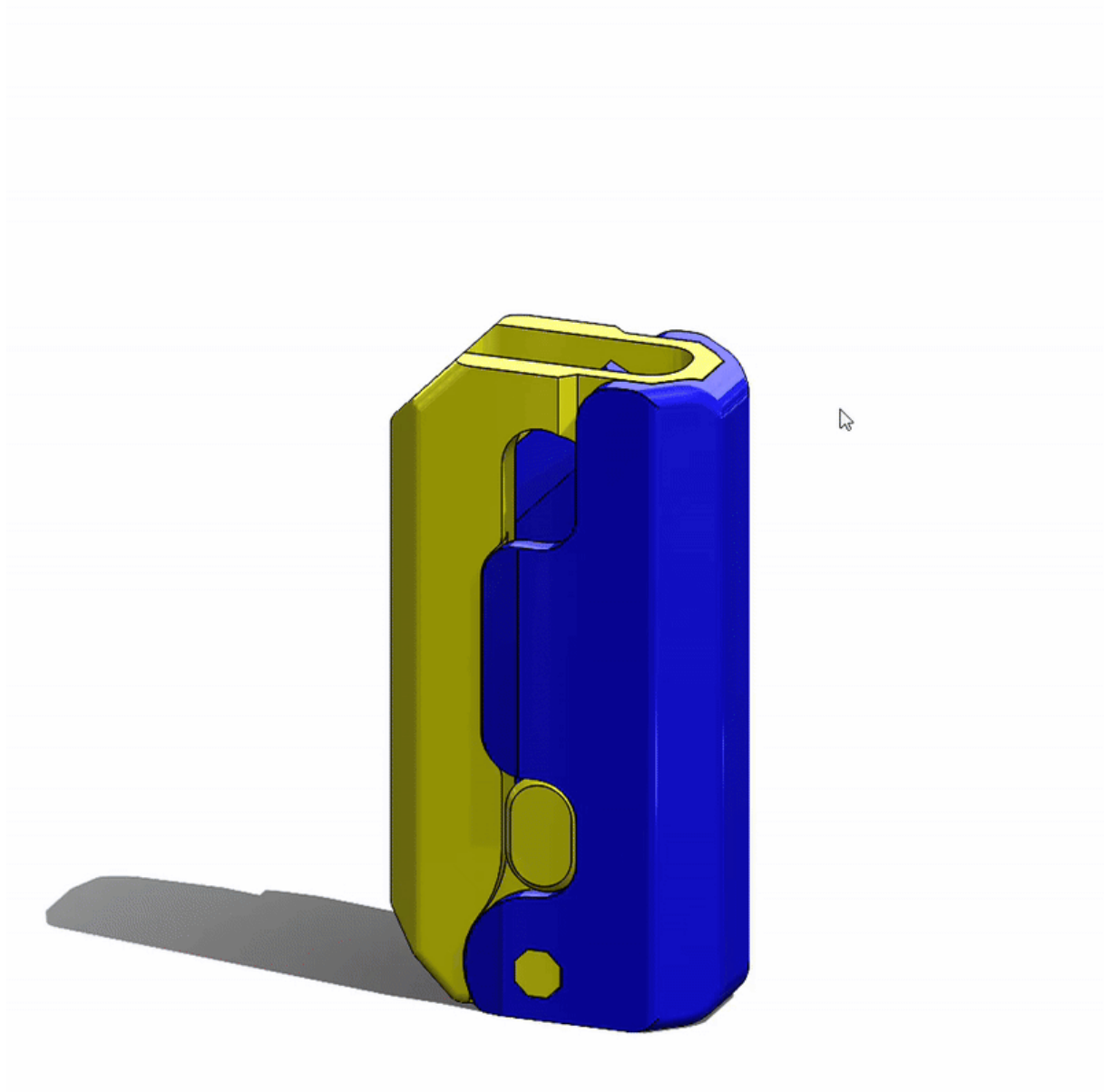
These are the overall dimensions of the set:



And these are its dimensions when the blade is out:



The operation of the knife is very simple, just tilt the guard to one side and slide the blade out:

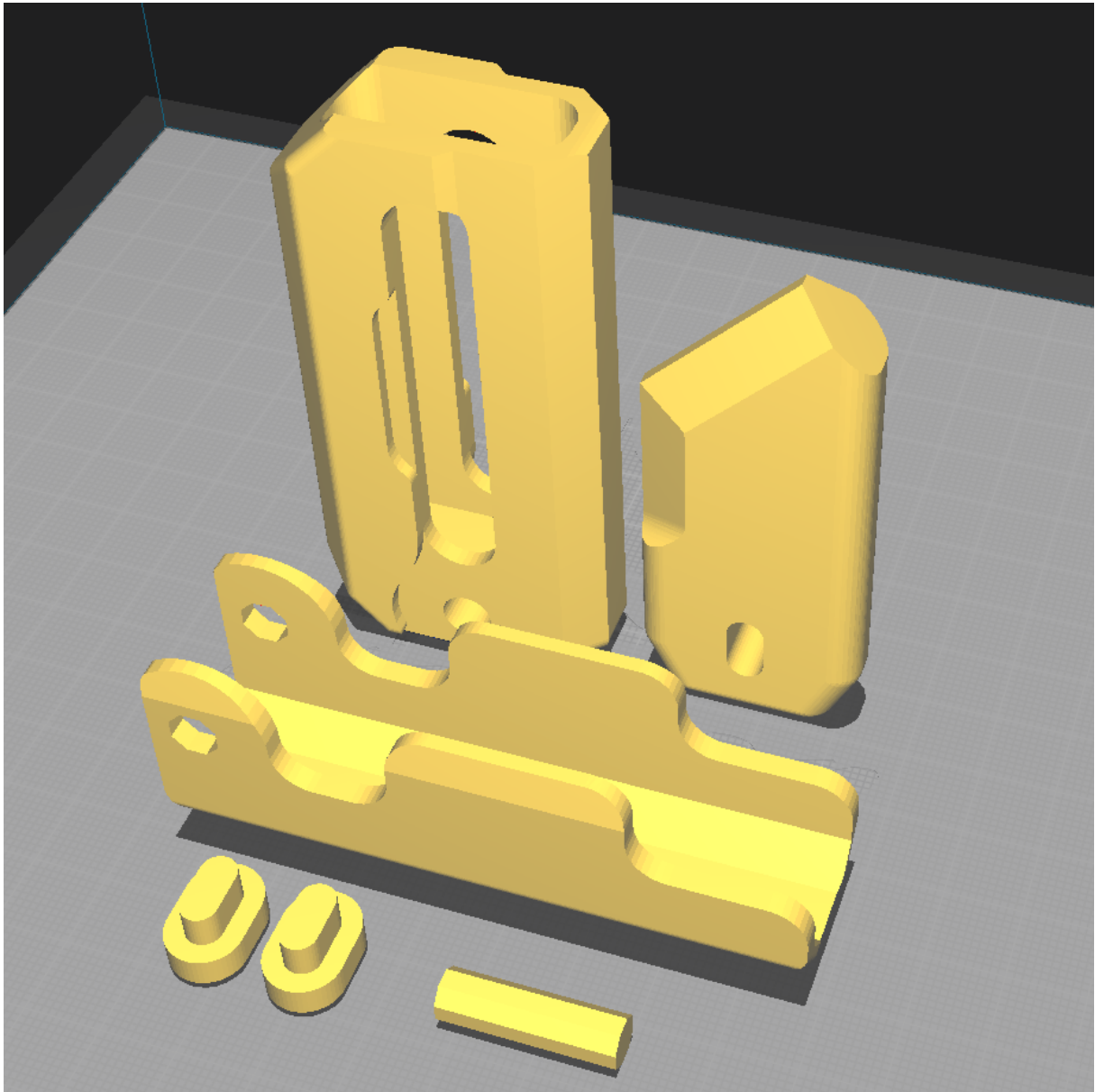


This sliding motion can be achieved with the thumb or by simply placing the knife downward letting gravity do its work, hence the name.

Print settings

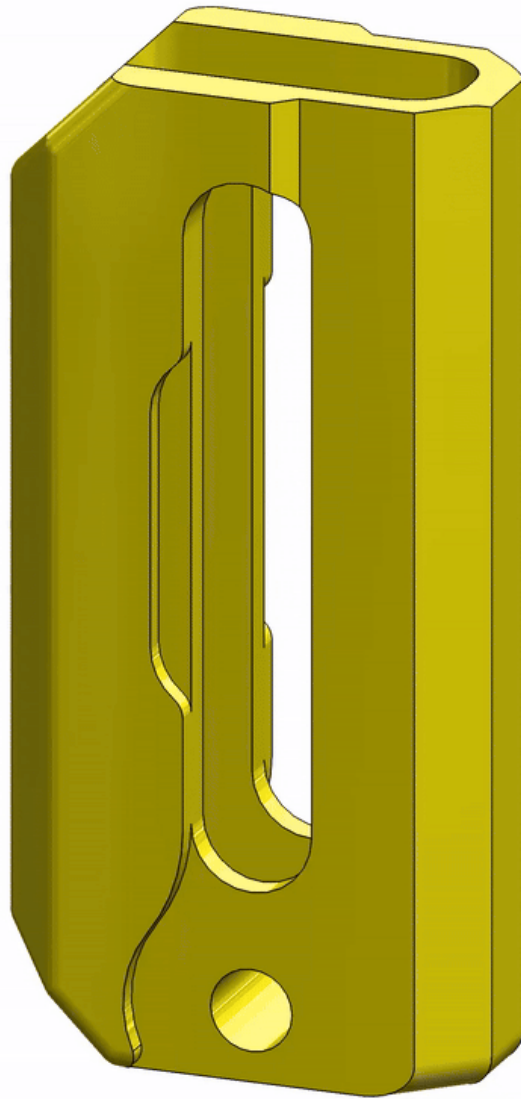
Everything printed in PLA and PETG, without rafts or supports, at 0.2 mm resolution and 15% infill.

As for the orientations, the following image shows the ideal orientation for each part:



Assembly

The assembly process is very simple as we can see in the following animation:



1. Insert the blade into the core.
2. Once the blade is inside, insert the stops into the blade.
3. Align the shaft hole on the guard with the one on the core.
4. Insert the shaft through the guard and the core at the same time.

If your printer is properly calibrated, everything should snap together without any problems.

However, if you want a more permanent mount, you can use a little of glue. Just be careful when gluing the shaft. You will only want to glue it to the guard, not the core, as this subassembly needs to be movable relative to the core. One drop on each side should be more than enough.

Model files



core.stl



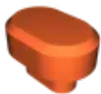
shaft.stl



blade.stl



guard.stl



stop.stl

Print files



ce3pro_fully-printable-gravity-knife-fidget-toy.gcode

🌀 PLA 🌀 0.40 mm ≡ 0.20 mm ⌚ 5.75 hrs 📊 36 g

📄 All the parts at once

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